HOLC Red-lining and Environmental Justice Issues in LA

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```
library(gt)
library(tidyverse)
library(sf)
library(here)
library(tmap)
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/home/jacobsloan/EDS223-hw-2/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.g
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/home/jacobsloan/EDS223-hw-2/data/mapping-inequality/mapping-inequality-
los-angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```
Reading layer `mapping-inequality-new-haven' from data source
  `/home/jacobsloan/EDS223-hw-2/data/mapping-inequality/mapping-inequality-
new-haven.json'
```

```
  using driver `GeoJSON'
Simple feature collection with 49 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -72.99321 ymin: 41.23109 xmax: -72.85227 ymax: 41.37502
Geodetic CRS:  WGS 84
```

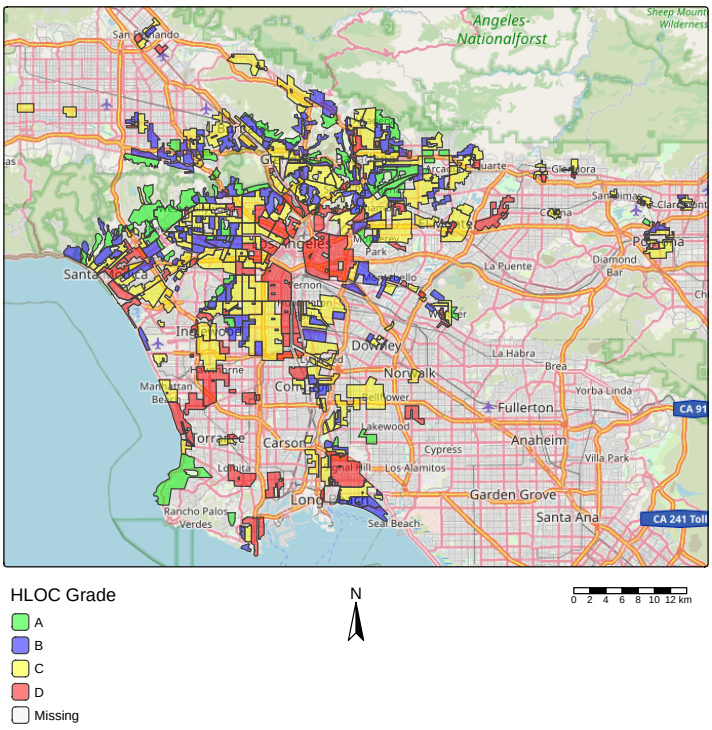
Part 1: Legacy of redlining in current environmental (in)justice

Create a map of historical redlined neighborhoods

```
# map of LA redlining
laejscreen <- ejscreen |>
    filter(CNTY_NAME == 'Los Angeles County')


tm_basemap('OpenStreetMap.DE') +
tm_shape(mapineqlos) +
    tm_polygons(
        fill = 'grade',
        fill_alpha = 0.5,
        fill.scale = tm_scale_categorical(values = c('green','blue','yellow','red')),
        fill.legend = tm_legend('HLOC Grade', frame = FALSE, position = tm_pos_out('center',
    )+
    tm_title(text = 'HOLC Ranking of Neighborhoods in LA') +
    tm_compass(type = 'arrow', position = tm_pos_out('center','bottom','center'))+
    tm_scalebar(position = tm_pos_out('center','bottom','right'))
```

HOLC Ranking of Neighborhoods in LA



Create table summarizing proportion of census block in each HOCL Grade

```
# Check CRS's before spatially joining
st_crs(mapineqlos)
```

```
Coordinate Reference System:
  User input: WGS 84
  wkt:
GEOGCRS["WGS 84",
    DATUM["World Geodetic System 1984",
        ELLIPSOID["WGS 84",6378137,298.257223563,
            LENGTHUNIT["metre",1]]],
    PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
    CS[ellipsoidal,2],
        AXIS["geodetic latitude (Lat)",north,
```

```
            ORDER[1],
            ANGLEUNIT["degree",0.0174532925199433]],
        AXIS["geodetic longitude (Lon)",east,
            ORDER[2],
            ANGLEUNIT["degree",0.0174532925199433]],
    ID["EPSG",4326]]
```

```
st_crs(ejscreen)
```

```
Coordinate Reference System:
  User input: WGS 84 / Pseudo-Mercator
  wkt:
PROJCRS["WGS 84 / Pseudo-Mercator",
    BASEGEOGCRS["WGS 84",
        ENSEMBLE["World Geodetic System 1984 ensemble",
            MEMBER["World Geodetic System 1984 (Transit)"],
            MEMBER["World Geodetic System 1984 (G730)"],
            MEMBER["World Geodetic System 1984 (G873)"],
            MEMBER["World Geodetic System 1984 (G1150)"],
            MEMBER["World Geodetic System 1984 (G1674)"],
            MEMBER["World Geodetic System 1984 (G1762)"],
            MEMBER["World Geodetic System 1984 (G2139)"],
            MEMBER["World Geodetic System 1984 (G2296)"],
            ELLIPSOID["WGS 84",6378137,298.257223563,
                LENGTHUNIT["metre",1]],
            ENSEMBLEACCURACY[2.0]],
        PRIMEM["Greenwich",0,
            ANGLEUNIT["degree",0.0174532925199433]],
        ID["EPSG",4326]],
    CONVERSION["Popular Visualisation Pseudo-Mercator",
        METHOD["Popular Visualisation Pseudo Mercator",
            ID["EPSG",1024]],
        PARAMETER["Latitude of natural origin",0,
            ANGLEUNIT["degree",0.0174532925199433],
            ID["EPSG",8801]],
        PARAMETER["Longitude of natural origin",0,
            ANGLEUNIT["degree",0.0174532925199433],
            ID["EPSG",8802]],
        PARAMETER["False easting",0,
            LENGTHUNIT["metre",1],
            ID["EPSG",8806]],
        PARAMETER["False northing",0,
```

```
            LENGTHUNIT["metre",1],
            ID["EPSG",8807]]],
    CS[Cartesian,2],
        AXIS["easting (X)",east,
            ORDER[1],
            LENGTHUNIT["metre",1]],
        AXIS["northing (Y)",north,
            ORDER[2],
            LENGTHUNIT["metre",1]],
    USAGE[
        SCOPE["Web mapping and visualisation."],
        AREA["World between 85.06°S and 85.06°N."],
        BBOX[-85.06,-180,85.06,180]],
    ID["EPSG",3857]]
```

```
# convert CRS's and then spatial join
mapineqlos<- st_transform(mapineqlos, crs = st_crs(ejscreen))
ejxred <- st_join(mapineqlos, ejscreen )

ejxred |>
    count(grade)
```

```
Simple feature collection with 5 features and 2 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -13203640 ymin: 3989343 xmax: -13102620 ymax: 4069679
Projected CRS: WGS 84 / Pseudo-Mercator
  grade    n                       geometry
1     A  449 MULTIPOLYGON (((-13193338 4...
2     B 1239 MULTIPOLYGON (((-13184185 4...
3     C 3058 MULTIPOLYGON (((-13188809 4...
4     D 1346 MULTIPOLYGON (((-13186948 4...
5  <NA>  296 MULTIPOLYGON (((-13164794 3...
```

```
# finding percent of blocks that are x Grade HOLC
gradeprop <- ejxred |>
    st_drop_geometry() |>
    select(c('grade','P_PM25','LOWINCPCT')) |>
    count(grade)
gradeprop$n <- gradeprop$n/nrow(ejxred)
gradeprop |>
    rename(proportion= n) |>
```

Proportion of Census Blocks in each HOCL Grade

grade	proportion
A	0.07028804
B	0.19395742
C	0.47871008
D	0.21070758
NA	0.04633688

```
    gt() |>
    tab_header('Proportion of Census Blocks in each HOCL Grade')
```

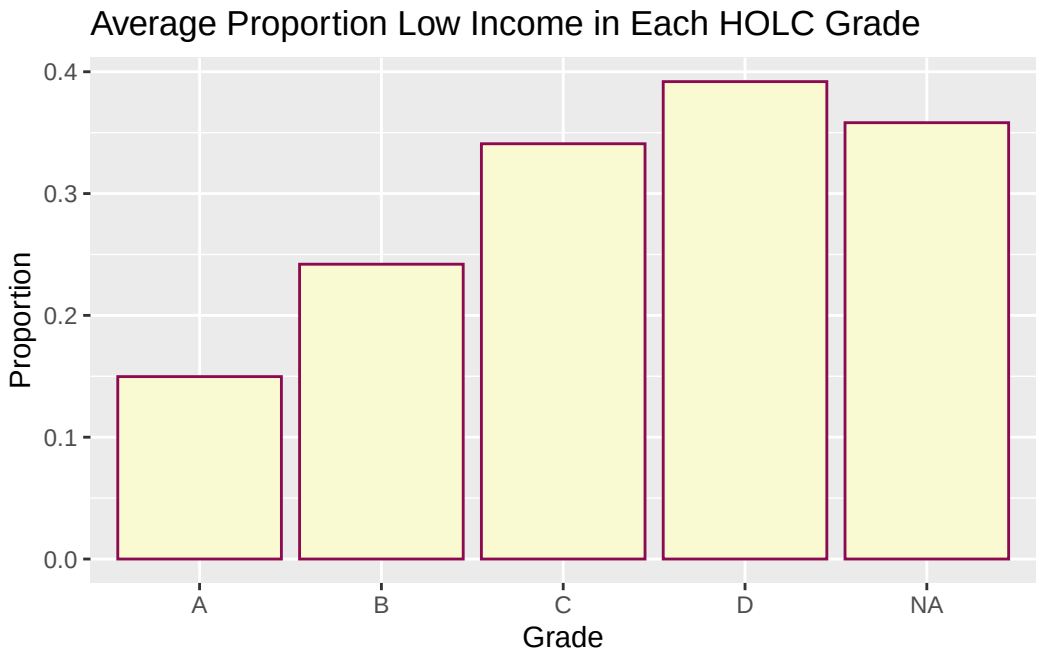
Create three visualizations summarizing current conditions

```
# Summarizing mean PM25 concentration, percent of people low income, and percent people with

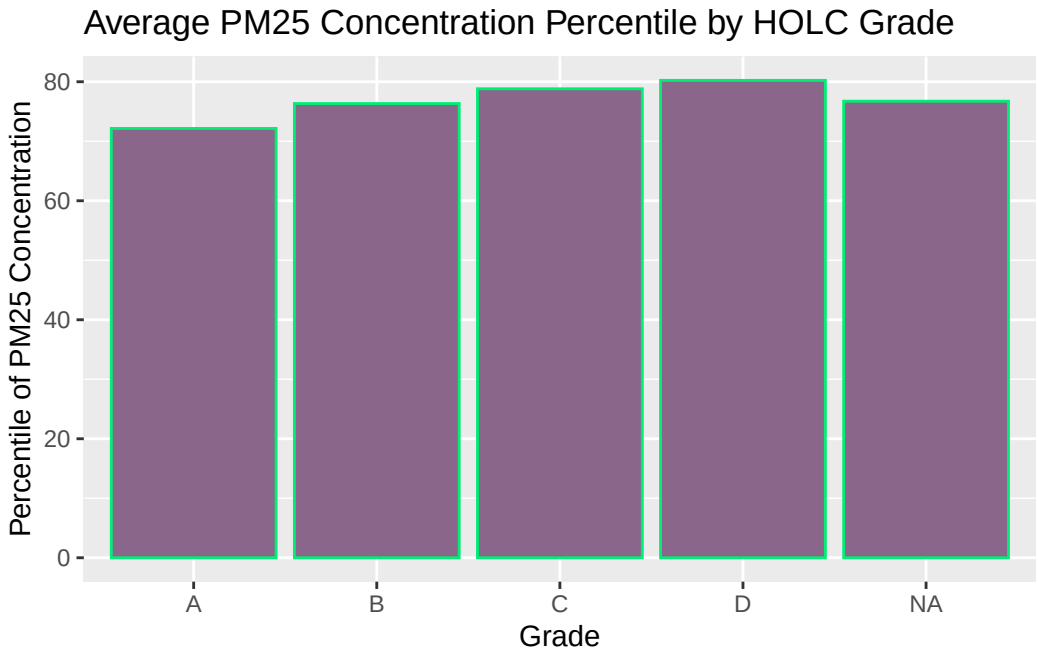
sumbygrade <- ejxred |>
    group_by(grade) |>
    summarise(meanlowinc = mean(LOWINCPCT, na.rm =TRUE), meanP_PM25 = mean(P_PM25, na.rm =TRU

# Plotting each EJSCREEN variable by HOCL grade

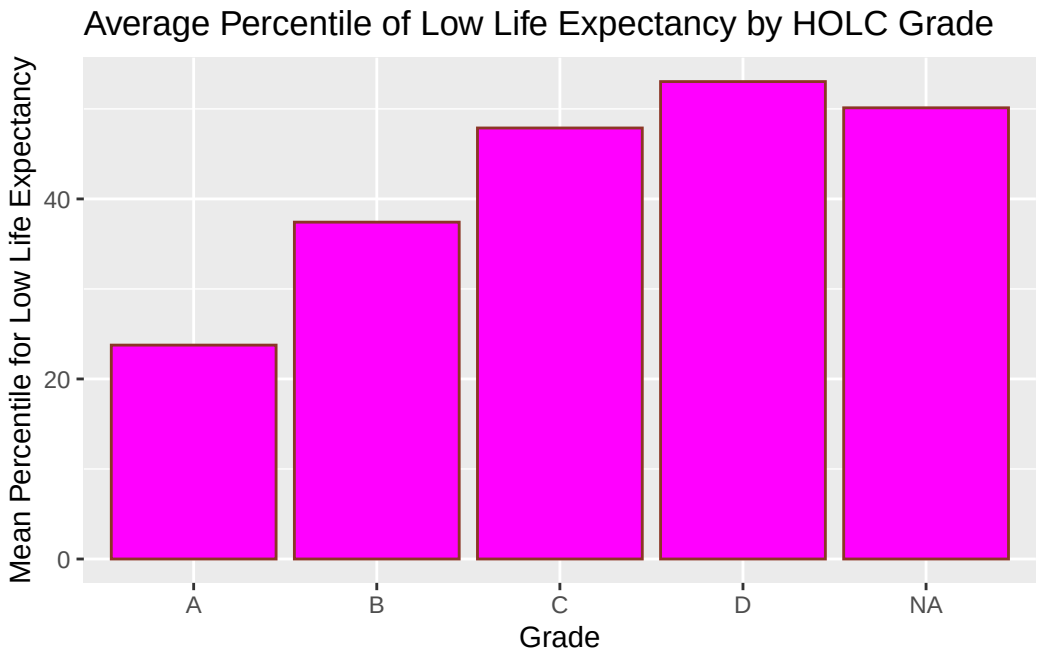
ggplot(sumbygrade, aes(x = grade, y = meanlowinc))+
    geom_bar(stat ='identity', fill = 'lightgoldenrodyellow', color = 'deeppink4') +
    labs(x = 'Grade',y = 'Proportion', title = 'Average Proportion Low Income in Each HOLC Gr
```




```
ggplot(sumbygrade, aes(x = grade, y = meanP_PM25))+
    geom_bar(stat ='identity', fill = 'plum4', color = 'springgreen2') +
    labs(x = 'Grade',y = 'Percentile of PM25 Concentration', title = 'Average PM25 Concentrat
```



```
ggplot(sumbygrade, aes(x = grade, y = meanlowlifexpect))+
    geom_bar(stat ='identity', color = 'tomato4', fill = 'magenta1') +
    labs(x = 'Grade',y = 'Mean Percentile for Low Life Expectancy', title = 'Average Percenti
```



Write a brief paragraph reflecting on these results

Visually, it seems that there is a relationship between the HOLC grade of a census block and % of people deemed low income, PM25, and percent of low life expectancy. The relationship is less obvious for PM25, which makes sense since air flow is less defined by neighborhood boundaries.

Part 2: Legacy of redlining in pedstrian and bike accidents

Figure summarizing the percent of observations within redlined neighborhoods within each HOLC grade

```
# Filter crash data to 2020-2025
accidentyearfilter <- crashes |>
    filter(ACCIDENT_YEAR >= 2020)
```

```r
# Check which columns have more NA's x/y or lat/long

sum(is.na(accidentyearfilter$POINT_X))
```

```
[1] 9484
```

```r
sum(is.na(accidentyearfilter$LATITUDE))
```

```
[1] 97350
```

```r
# Way more NAs in lat/lon, lets use x y instead

# Get ride of NA's in coordinates
accidentyearfilter <- accidentyearfilter |>
    filter(POINT_X != 'NA')
# Convert character columns to geometries

accidentyearfilter <- st_as_sf(accidentyearfilter, coords = c('POINT_X', 'POINT_Y'), crs = 4
```

```r
# Joining accident df with HOLC in LA DF

# Check CRS's before spatially joining
if (st_crs(accidentyearfilter) == st_crs(mapineqlos)) {
    print('It is all good!')
} else {
    warning('It is not good the CRS do not match')
}


# convert CRS's and then spatial join
accidentyearfilter<- st_transform(accidentyearfilter, crs = st_crs(mapineqlos))

# Join accidents with HOLC in LA
accxHOLC <- st_join(mapineqlos, accidentyearfilter)
```

```r
# Visualize proportion of pedestrian accidents and fatalities by HOLC Grade
proppedHOLC <- accxHOLC |>
    st_drop_geometry() |>
    filter(PEDESTRIAN_ACCIDENT == 'Y') |>
```

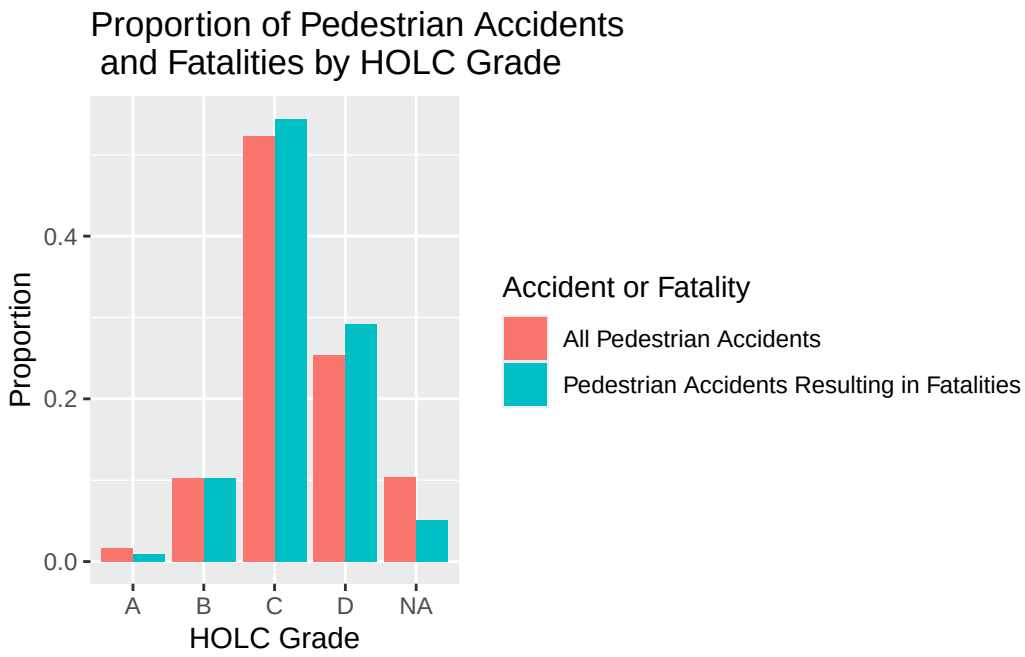
```
    count(grade)

proppedfata <- accxHOLC |>
    st_drop_geometry() |>
    filter(COUNT_PED_KILLED > 0) |>
    count(grade)


proppedHOLC$propacc <- proppedHOLC$n/sum(accxHOLC$PEDESTRIAN_ACCIDENT == 'Y', na.rm = TRUE)
proppedHOLC$propfata <- proppedfata$n/sum(accxHOLC$COUNT_PED_KILLED > 0, na.rm =TRUE)

proppedHOLC <- pivot_longer(proppedHOLC, cols = c('propfata','propacc'), names_to = 'aorf', v

ggplot(proppedHOLC, aes(fill = aorf, x = grade, y = prop)) +
    geom_bar(position = 'dodge',stat = 'identity') +
    labs(title = 'Proportion of Pedestrian Accidents \n and Fatalities by HOLC Grade', x = 'H
    scale_fill_discrete(labels = c('All Pedestrian Accidents','Pedestrian Accidents Resulting
```




```
# Visualize proportion of bicycle accidnets by HOLC Grade

propbikeHOLC <- accxHOLC |>
    st_drop_geometry() |>
```

```
    filter(BICYCLE_ACCIDENT == 'Y') |>
    count(grade)

propbikefata <- accxHOLC |>
    st_drop_geometry() |>
    filter(COUNT_BICYCLIST_KILLED > 0) |>
    count(grade)

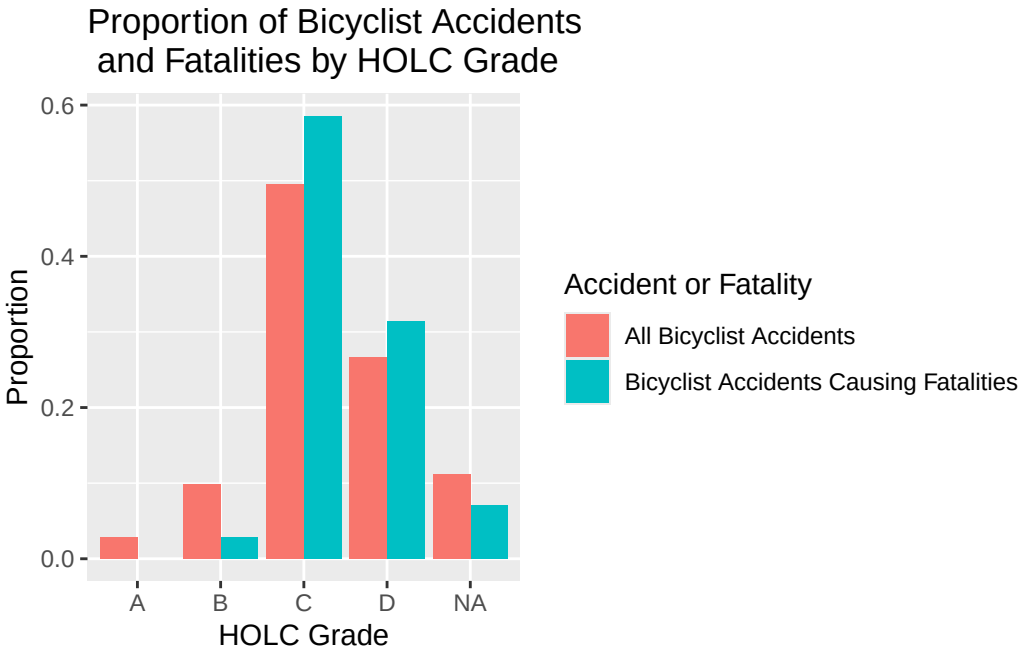

propbikeHOLC$propacc <- propbikeHOLC$n/sum(accxHOLC$BICYCLE_ACCIDENT == 'Y', na.rm = TRUE)

bikedeaths <- accxHOLC |>
    group_by(grade) |>
    filter(COUNT_BICYCLIST_KILLED > 0) |>
    count(grade) |>
    mutate(propdeath = n/sum(n))

propbikeHOLC <- left_join(propbikeHOLC, bikedeaths, by = 'grade')

propbikeHOLC <- pivot_longer(propbikeHOLC, cols = c('propdeath','propacc'), names_to = 'aorf

ggplot(propbikeHOLC, aes(fill = aorf, x = grade, y = prop)) +
    geom_bar(position = 'dodge',stat = 'identity') +
    labs(title = 'Proportion of Bicyclist Accidents \n and Fatalities by HOLC Grade', x = 'HO
    scale_fill_discrete(labels = c('All Bicyclist Accidents','Bicyclist Accidents Causing Fat
```


Proportion of Bicyclist Accidents
and Fatalities by HOLC Grade
Proportion
0.6
0.4
0.2
0.0
A
B
C
D
NA
HOLC Grade
Accident or Fatality
All Bicyclist Accidents
Bicyclist Accidents Causing Fatalities